WLM Foundation Series
Paper 4
Structure–Appearance Series

The Uniqueness and Unskippability of Invariant Structure:

Recursion as the Unique Generator of Structure

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Abstract

Paper 1 established 0 → ε. Paper 2 established ε → {+ε, −ε}. Paper 3 established {+ε, −ε} → recursion. This paper establishes the fourth forced link: recursion → invariant structure. Under R1 and R2, recursion necessarily generates invariant structure; this structure is unique at the minimal level; and no invariant structure exists without recursion. Structure is unskippable in any chain leading to appearance.

This completes the transition from generative dynamics to stable structure, setting up the final step in Paper 5: structure → appearance.

Epistemic status. *All theorems follow from axioms and results in Papers 0–3. No new axioms are introduced. This paper adds the fourth partial result toward Conjecture U1.*

0 → ε (P1) → ±ε (P2) → recursion (P3) → structure (P4) → appearance (P5)

0. Framework and Inherited Results

Paper 4 operates entirely within the axiom system of Papers 0–3. No new axioms are introduced.

Axiom S0 (Symmetry Closure). *Any state of perfect indistinction is a fixed point of every generative rule.*

Axiom D0 (Generative Orientation and Reversal). *(a) Any $x \nsim_0 0$ carries orientation label $\theta(x)$. (b) There exists a generator realizing orientation reversal on ε.*

Axiom R1 (Composability). *Generative rules are closed under composition; iterates f^n are defined for all $n \geq 1$.*

Axiom R2 (Recursive Orientation Preservation). *Directional flips persist under all iterates.*

Axiom V-WF (Well-Founded Valuation). *The set $\{v(x) : v(x) > v(0)\}$ is well-founded; every nonempty subset has a minimal element.*

Key inherited results used without re-proof:

- Paper 0, Theorem 5: Any non-collapsing recursive orbit generates at least one invariant structure (periodic cycle or compact closure by Bolzano-Weierstrass).
- Paper 0, Lemma 4c: Every non-constant orbit in a non-collapsing chain alternates $\pm\varepsilon$.
- Paper 3, Theorem 4.3: Recursion is unskippable for any chain reaching non-trivial dynamics.
- Paper 3, Theorem 4.1: Recursion under R1 + R2 produces the minimal non-trivial invariant sets.
- Paper 2, Theorems 4.2 and 4.3: Duality is the unique generator of recursion and is unskippable.
- Paper 1, valuation $v(\cdot)$ and minimality of ε.

Remark (Structural Parallelism). *Paper 4 mirrors the architecture of Papers 1–3: Section 1 defines the minimal structure of the new layer; Section 2 shows it is forced by recursion; Section 3 classifies all collapse modes; Section 4 proves invariant stability, necessity, and unskippability.*

1. Structure Space

Definition 1.1 (Invariant Set). *A set $S \subseteq X$ is an invariant structure if there exists a recursive generator f such that $f(S) \subseteq S$. (We use the inclusive form $f(S) \subseteq S$ rather than $f(S) = S$ to accommodate compact closures; strict equality holds for periodic cycles.)*

1.1 Minimal Structure

Theorem 1.1 (Minimal Structure). *Under R1 and R2, the minimal non-trivial invariant structure is the closure of the persistent ±ε alternating orbit.*

Proof. By Paper 0 Theorem 5, any non-collapsing recursive orbit generates at least one invariant set — either a periodic cycle or a compact closure. By Paper 3, the only persistent non-trivial orbits at the minimal valuation level are those alternating ±ε under R2. By Paper 2, ±ε is the unique structure at this level. Therefore the smallest invariant structure is precisely the closure of this alternating orbit. □

1.2 Structure Irreducibility

Theorem 1.2 (Structure Irreducibility). *No non-recursive or non-alternating mechanism can produce invariant structure.*

Proof. Invariant structure requires $f(S) \subseteq S$ for some recursive f (Definition 1.1). This closure on a non-trivial set requires both recursion (R1, for the iterates to be defined) and duality (Paper 2, for the alternation that the orbit exhibits). Without either, only the collapse modes of Paper 3 remain: static collapse (no R1) or divergent collapse (no closure). Invariant structure is therefore irreducible to any simpler mechanism. □

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2. Structure Forced by Recursion

2.1 Recursion Generates Structure

Theorem 2.1 (Recursion Requires Structure). *Any non-collapsing recursive orbit necessarily generates invariant structure.*

Proof. Direct from Paper 0 Theorem 5: if an orbit is non-collapsing and recursive (R1 gives defined iterates for all n), it is either eventually periodic — generating an invariant cycle — or infinite and bounded — generating a compact invariant closure by Bolzano-Weierstrass. In both cases, an invariant structure S with $f(S) \subseteq S$ is produced. □

2.2 Uniqueness at Minimal Level

Theorem 2.2 (Structure Uniqueness at Minimal Level). *At the minimal valuation level, recursion generates a unique invariant structure up to observational equivalence.*

Proof. By Paper 1, ε is unique up to observational equivalence. By Paper 2, ±ε is the unique dual pair at valuation v(ε). By Paper 3, recursion on ±ε is the unique engine. The only persistent alternating orbit at the minimal level is therefore the one generated from {+ε, −ε} under the unique recursive engine. Its closure is the unique minimal invariant structure.

Any other candidate at the same valuation level is observationally equivalent by the valuation property (Paper 1, Definition 0.4). □

3. Collapse of Non-Structural Chains

The following collapse modes exhaust all possible failures of the structure layer. Every generative chain that fails to produce invariant structure falls into one of these four categories.

3.1 Oscillation Collapse

Theorem 3.1 (Oscillation Collapse). *If recursion is present but produces no invariant closure, the system collapses to persistent ±ε oscillation without higher structure.*

Proof. A recursive orbit that alternates ±ε (Lemma 4c) but does not produce a closed invariant set remains a pure alternating sequence. Without a fixed set S satisfying $f(S) \subseteq S$, no stable structure forms: the orbit oscillates indefinitely but generates no invariant. This is distinct from static and divergent collapse — the dynamics are non-trivial, but no structure crystallizes. □

3.2 Divergent Collapse

Theorem 3.2 (Divergent Collapse). *If recursion is replaced by non-closed iteration, no invariant structure forms.*

Proof. Without R1, orbits are either undefined for large n or escape bounded valuation regions. By Paper 0 Theorem 5, invariant structure requires a bounded non-collapsing orbit (so that Bolzano-Weierstrass applies). Unbounded iteration produces no compact invariant set. □

Figure 1: Collapse modes of the structure layer.

Collapse Mode	Trigger Condition	Consequence	Reference
Static Collapse	No recursion (R1 absent)	No invariants; trivial dynamics	Paper 3, Thm 3.1
Divergent Collapse	Non-closed iteration	Unbounded orbits; no invariants	Thm 3.2
Oscillation Collapse	Recursion present; no invariant closure	Pure ±ε alternation; no stable structure	Thm 3.1
Duality-Only Collapse	±ε without recursion	Trivial dynamics	Paper 2 / Paper 3

4. Structure as Generative Invariant

4.1 Invariant Stability

Theorem 4.1 (Invariant Stability). *Once formed, invariant structure S is preserved under all further iterates of the generating rule: $f^n(S) \subseteq S$ for all $n \geq 1$.*

Proof. By definition, $f(S) \subseteq S$. By R1, $f^2(S) = f(f(S)) \subseteq f(S) \subseteq S$. By induction, $f^n(S) \subseteq S$ for all $n \geq 1$. Invariance is preserved under all iterates of the generating rule without requiring any additional assumption about other generators. □

Remark. *Theorem 4.1 is stated for the generating rule f, not for arbitrary composed generators $H \circ f$. If H also preserves S (i.e., $H(S) \subseteq S$), then $(H \circ f)(S) \subseteq S$ follows. But $H(S) \subseteq S$ is an additional condition on H, not a consequence of R1 alone. The theorem as stated — stability under iterates of f — is what follows strictly from R1 and the definition of invariance.*

4.2 Necessity for Appearance

Theorem 4.2 (No Appearance Without Structure). *Appearance via P0 requires stable invariants; invariants require recursion. Therefore structure is necessary for appearance.*

Proof. By Paper 0 Theorem 6 (Projection Necessity) and Axiom P0, appearance requires a projection $P: D \to S$ from stable invariants. Without invariants, there is nothing for P to act on: a projection from an empty or undefined domain has no observational content. By Theorem 1.2 (Structure Irreducibility), invariants require recursion. Therefore appearance requires recursion through structure. □

4.3 Structure is Unskippable

Theorem 4.3 (Structure is Unskippable). *Any generative chain realizing appearance must pass through the invariant structure layer. No chain produces appearance while bypassing structure.*

Proof. Contrapositive of Theorems 3.1, 3.2, and 4.2: (i) oscillation collapse (Thm 3.1) produces no invariant, hence no appearance; (ii) divergent collapse (Thm 3.2) produces no invariant, hence no appearance; (iii) by Thm 4.2, appearance requires invariants directly. In all cases of structure absence, appearance is impossible. Therefore any chain reaching appearance must have passed through invariant structure. This is the structure analogue of Paper 1 Thm 3.3 (ε unskippable), Paper 2 Thm 4.3 ($\pm\varepsilon$ unskippable), and Paper 3 Thm 4.3 (recursion unskippable). It constitutes the fourth partial result for Conjecture U1. □

Remark (Unskippability Chain). *ε (Paper 1) → $\pm\varepsilon$ (Paper 2) → recursion (Paper 3) → structure (this paper). Four layers confirmed. The chain is locked through four unskippable transitions. Paper 5 will complete the chain with structure → appearance.*

5. Structural Summary

Recursion (R1 + R2) on the dual pair {±ε} necessarily generates invariant structure (Paper 0 Thm 5). This structure is unique at the minimal level. It is stable under all further iterates of the generating rule. Appearance requires it. It cannot be skipped.

Figure 1: The Structure Layer

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        {+ε, −ε}
        |   (R1 + R2)
            ▼
        Recursion
            |
            ▼
   Invariant Structure
    (minimal stable set)
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Figure 2: Full Generative Chain

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0 ──S0──→ ε ──D0──→ {±ε} ──R1+R2──→ Recursion ──→ Structure ──P0──→
                         Appearance
                                                    |
                                              Collapse Modes
                         (Static / Divergent / Oscillation / Duality-
                                          only)
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Figure 3: Parallelism Across Papers 1–4

Paper	Forced Step	Key Property	Unskippable Result
1	0 → ε	Minimal unique offset	Thm 3.3
2	ε → {±ε}	Binary orientation space	Thm 4.3
3	{±ε} → Recursion	Unique generative engine	Thm 4.3
4	Recursion → Structure	Unique minimal invariant	Thm 4.3

Each paper adds one confirmed unskippable layer. The four results together constitute the current state of Conjecture U1: the chain through four layers is unique and forced. Paper 5 will add the fifth and final layer (structure → appearance) and complete the chain.

Conclusion

Paper 4 establishes the fourth forced link of the WLM generative chain: recursion → invariant structure. Any non-collapsing recursive orbit necessarily generates invariant structure (Theorem 2.1). This structure is unique at the minimal valuation level (Theorem 2.2). It is stable under all further iterates of its generating rule (Theorem 4.1). Appearance requires it (Theorem 4.2). And no chain can bypass it (Theorem 4.3).

The four unskippability results confirmed so far: ε (Paper 1 Thm 3.3); ±ε (Paper 2 Thm 4.3); recursion (Paper 3 Thm 4.3); structure (Thm 4.3, this paper). Paper 5 will establish structure → appearance as the fifth and final forced step, completing Conjecture U1.

References

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Bolzano-Weierstrass theorem. — Used in Paper 0 Theorem 5, inherited here: bounded infinite orbits have limit points, providing compact invariant closures.

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